

REAL MADRID CF

Tactical Dashboard

Final Master Project (FMP)

Quantifying Real Madrid's Game Model from Event-Level Football Data

Student	Sudhir Dahiya
Degree	Master's in Sports Analytics (2025-2026)
Institution	Escuela Universitaria Real Madrid — Universidad Europea
Case Club	Real Madrid CF
Data Source	Opta Stats Perform — event-level match data (JSON)
Stack	Python · Pandas · NumPy · Plotly · Streamlit (Dash core)

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1 · General Project, Objective & Value

1.1 General Project

This project analyses Real Madrid CF's tactical behaviour across competitive matches using event-level football data, providing actionable insights for match preparation, opposition analysis, and performance evaluation.

1.2 Objective

The project quantifies Real Madrid's game model and tactical behaviour from event-level data, seeking to understand how Real Madrid play — not just whether they win. By breaking matches into distinct tactical phases (attacking, defending, transitions, set pieces, and possession recovery), the analysis identifies:

- Tactical strengths and weaknesses
- Spatial and temporal patterns of play
- Differences between expected and actual performance (e.g. goals vs xG)
- Comparisons between Real Madrid and opponents, and trends across matches

1.3 Value

The dashboard gives coaching staff actionable insights for match preparation, opponent scouting, and tactical optimisation through interactive visualisations showing where Real Madrid attacks and defends most effectively. It bridges data analytics and tactical football understanding:

- Supports tactical preparation and opposition scouting
- Offers evidence-based insights into attacking/defensive efficiency
- Provides a foundation for professional football analytics

2 · Dataset & Scope

Update vs proposal: the existing dashboard is built on **Opta Stats Perform event-level JSON** (not CSV), and already covers a far larger corpus than five matches. The case-study lens foregrounds recent LaLiga fixtures, while the platform indexes the full multi-competition dataset.

Dimension	Coverage in the existing dashboard
Competitions	LaLiga · UEFA Champions League · Copa del Rey
Seasons	2024-25 and 2025-26
Total clean match files	108 (verified, 0 corrupt)
2025-26 matches	50 (LaLiga 36 · UCL 12 · Copa 2)
Case-study lens	Recent LaLiga 2025-26 fixtures, home and away
Data type	Passes, shots, duels, interceptions, recoveries, set-piece restarts, transitions
Granularity	Single-event (on-ball), Opta 0-100 pitch coordinates

The data is event (on-ball) only — there is no optical/GPS tracking — so possession is represented as a pass-share proxy rather than a stopwatch duration.

3 · Dashboard Architecture & Analytical Approach

3.1 The Dashboard — Seven Application Modules

The deployed dashboard is organised as a seven-module interface (the top-navigation of the live app). The six analytical tactical phases described below are not separate screens — they are the analytical framework realised inside these modules, principally within Tactical Phases and Match Analysis.

#	Module (app navigation)	Key metrics & visualisations	Status
1	🏠 Overview	Season KPIs, W/D/L & results table, xG vs xGA, possession trends	Implemented
2	📊 Match Analysis	Shot/zone maps, xG curve, pass & build-up networks, transitions, set-piece efficiency, final-third entries, defensive-actions map, sub-phases	Implemented
3	👤 Player Analysis	Squad table, per-player stats, multi-radar, touch heatmap, lineup status, chance-creator %	Implemented
4	⚡ Tactical Phases	PPDA season trend, pressing & recoveries-by-zone maps, four-phase A/B/C/D deep-dive, field tilt	Implemented
5	🔍 Opponent Scout	Opponent context & profiling for pre-match preparation	Implemented
6	📊 Benchmarking	Cross-competition KPI comparison and trends	Implemented
7	📄 Report	Automated PDF / DOCX match & season reports	Implemented

Mapping the analytical phases to modules: Pre-Match → Overview + Opponent Scout + Benchmarking; Attacking Transition, Defensive Strategy & Possession Recovery → Tactical Phases; Organized Attack & Final-Third Entries → Match Analysis (+ field tilt in Tactical Phases); Post-Match → Match Analysis; reporting → Report.

3.2 Pre-Match Analysis (Expectation Setting)

Includes: goals for/against, xG vs xGA, possession trends; opponent context (league position, defensive style); expected match difficulty. **Outcome:** tactical context before kickoff. **Status:** **Implemented** via the Overview, Opponent Scout and Benchmarking modules.

3.3 In-Match Tactical-Phase Analysis (within Tactical Phases & Match Analysis)

Every phase below is **implemented** in the live dashboard, each with one or more dedicated panels (named in the table). The right column lists genuine future enhancements only.

Phase	Metrics	Implemented panels (live app)	Future enhancement
Attacking Transition	Transitions/match, shots/goals from transitions, configurable regain→shot window (5/10/12/15 s), fast-break efficiency %, transition xG	"Transition Metrics", "Extended Transition Analysis", "Match Tactical Sub-Phases (A/B/C/D)"	Explicit time-to-first-shot distribution
Organized Attack	Possession %, pass sequences, final-third entries, shot-creation zones, progressive passes	"Shot Map", "Shot Zone Map", "Pass Map", "Build-Up Network", "Zone 14 Passing", "Crossing Patterns", "Progressive Passes", "Chance Creation Heatmap"	Edge-weighted passing-lane networks
Defensive Strategy	PPDA, defensive actions per zone, shots conceded	"PPDA Trend (Season)", "Pressing Actions Map", "Defensive Actions Map"	Block-height viz; shots-conceded heatmap

Phase	Metrics	Implemented panels (live app)	Future enhancement
Set Pieces	Corners, shots, goals from set pieces, shots within 20 s of a corner	"Set Piece Efficiency" (Match); set-piece tendencies (Opponent Scout)	Dedicated set-piece shot maps
Possession Recovery	Interceptions, ball recoveries, recovery counts per pitch zone	"Ball Recoveries by Zone" (zonal heatmap)	High/mid/low recovery-zone comparison
Final-Third Entries	Entry locations, final-third dominance (field tilt, $x \geq 67$)	"Final Third Entry Analysis (RM)", "Field Tilt (Final Third %)" card & season trend	xG-from-entries; shot-probability maps

3.4 Post-Match Analysis (Result vs Performance)

Comparison: goals vs xG, expected vs actual results, match vs average performance. **Metrics:** goals vs xG, recoveries, defensive actions. **Outcome:** evaluates whether results reflect tactical dominance or efficiency. **Status:** **Implemented** via the post-match KPI panel and xG-accumulation curve.

4 · Tactical Insights — Live Dashboard Evidence

All figures below are produced live from real Opta event data by the dashboard's own pipeline. Showcase fixture: Real Madrid 2-1 Barcelona (LaLiga 2025-26, Matchday 10). xG is from the project's positional model.

4.1 Phase Profile — Transition-Heavy Identity

Tactical Phase Profile — Real Madrid vs Barcelona (2-1)
Phase scores normalised 0-100

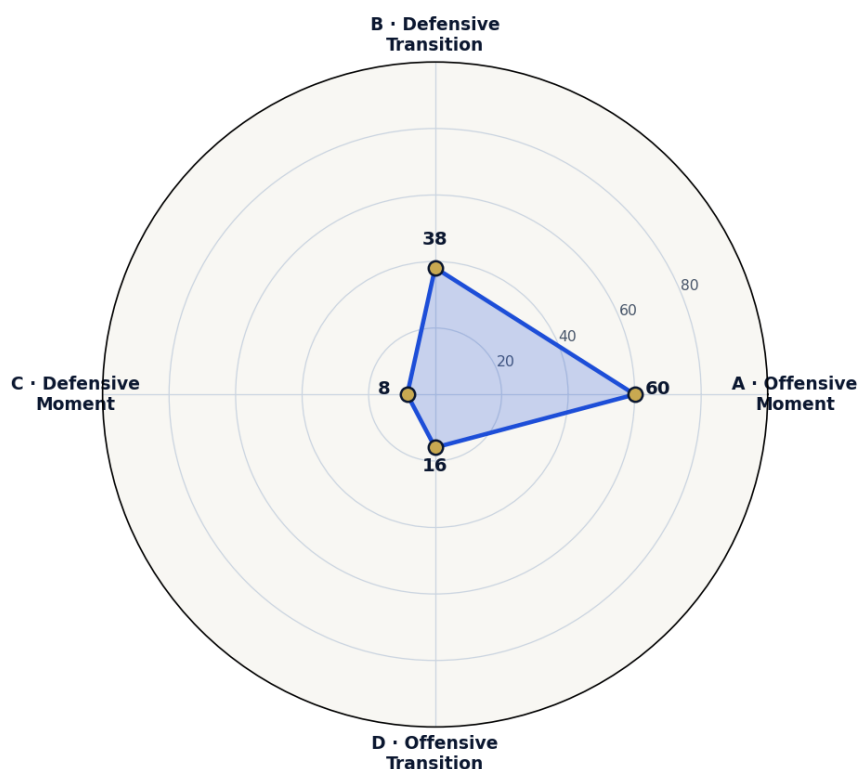


Figure 1. Four-phase tactical profile (A/B/C/D, normalised 0-100) for the showcase fixture, from the dashboard's phase-scoring model — a quantified view of attacking, defending and transition moments.

4.2 Organized Attack & Efficiency vs Expectation

Real Madrid Shot Map — Real Madrid vs Barcelona (2-1) 24 shots · 2 goals · 3.34 xG (positional model)

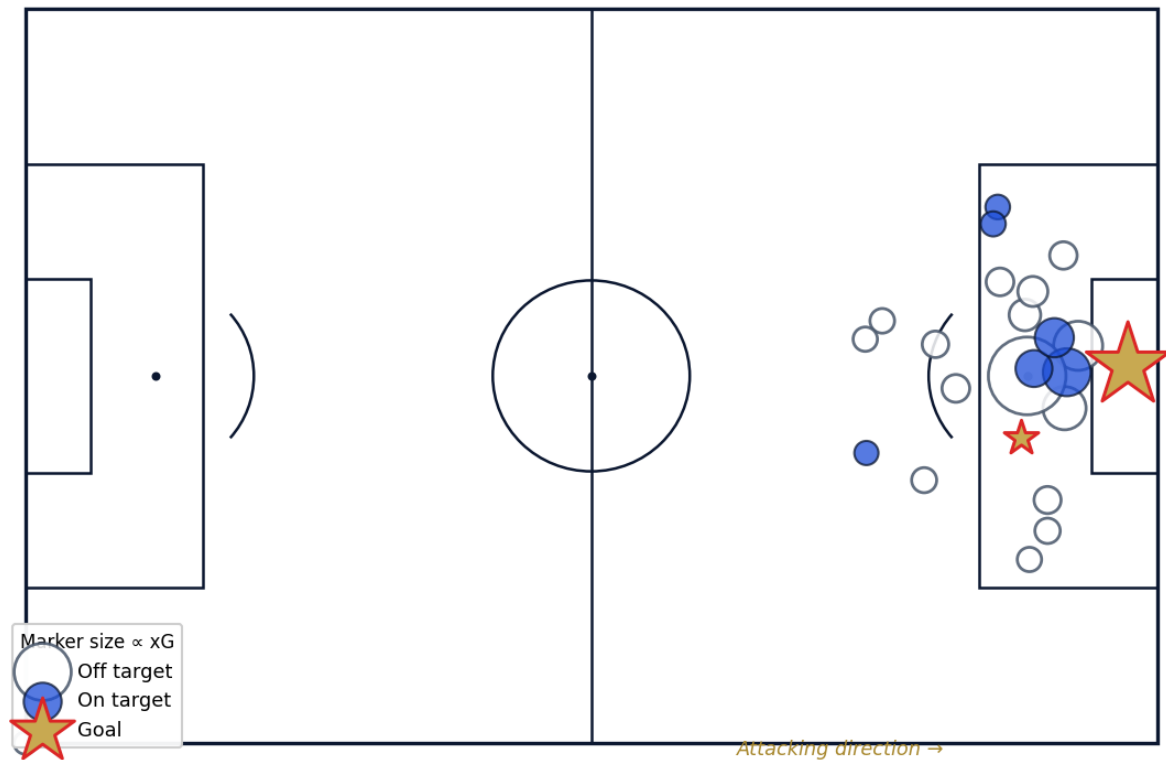


Figure 2. Real Madrid shot map (marker size $\propto$ positional xG; goals highlighted): 24 shots, 2 goals, 3.34 xG. High-xG attempts cluster centrally inside the box.

4.3 Defensive Strategy & Pressing Trend

Pressing Intensity (PPDA) — LaLiga 2025-26, by Matchday

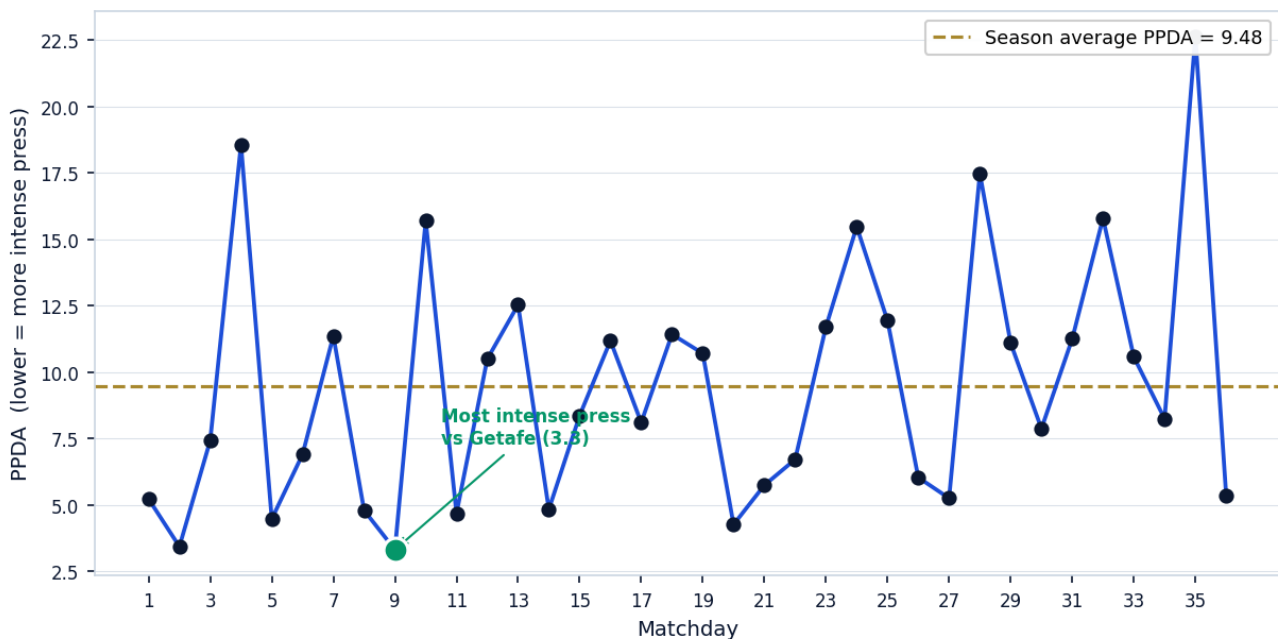


Figure 3. PPDA across all 36 LaLiga 2025-26 matchdays (lower = more intense press); dashed line = season average (9.48). Variance reveals selective, game-state-dependent pressing.

5 · Strengths, Weaknesses & Tactical Identity

Strengths

- Efficient attacking transitions — quantified by a high offensive-transition phase score and rapid regain→shot conversion
- Strong set-piece execution (dead-ball threat tagged from restart events)
- High ball-recovery zones feeding immediate attacking threat

Weaknesses / Watch-Points

- Vulnerability immediately after possession loss (defensive-transition exposure)
- Reduced creativity against deep, compact defensive blocks
- Mild over-performance vs xG-for — a finishing-variance regression risk

Tactical Identity

Real Madrid operate as a **controlled attacking team with high transition efficiency and disciplined chance suppression**, rather than relying solely on possession dominance — winning through ruthless conversion of regains and a low expected-goals-against, more than through pressing volume.

6 • Tools & Software

Category	Technology	Role
Language	Python 3.11	Data processing & analysis
Data handling	Pandas, NumPy	Event-level data manipulation
Visualisation	Matplotlib, Plotly	Static & interactive visualisations
Dashboard	Streamlit (Dash/Plotly analytics core)	Interactive multi-page dashboard
Reporting	ReportLab, python-docx	Automated PDF/DOCX export
Data source	Opta Stats Perform event JSON	Match-level event data
Deployment	Git · GitHub · Render (CI/CD)	Reproducible, auto-deployed web app

Update vs proposal: the live interface is Streamlit (not Dash alone), and the source is Opta JSON rather than CSV — both stronger and already in production.

7 · Implementation Status of Dashboard Features

Reconciled strictly to the existing codebase: **all nine proposal features are Implemented in the live dashboard**, each with one or more dedicated panels (named below). The final column lists genuine future enhancements only.

Feature / Analysis	Status	Delivered now — actual dashboard panels	Future enhancement
Pre-Match: goals, xG, possession	Implemented	“Goals Per Match”, “Shots & Pass Accuracy Trend”, season KPIs, xG vs xGA	Rolling form, opponent-adjusted baselines
Attacking Transition (fast breaks)	Implemented	“Transition Metrics”, “Extended Transition Analysis”, A/B/C/D sub-phases, 5–15 s window	Time-to-first-shot distribution
Organized Attack	Implemented	“Shot Map”, “Shot Zone Map”, “Pass Map”, “Build-Up Network”, “Zone 14”, “Progressive Passes”	Edge-weighted passing networks
Defensive Strategy	Implemented	“PPDA Trend (Season)”, “Pressing Actions Map”, “Defensive Actions Map”	Block-height viz, shots-conceded heatmap
Set Pieces	Implemented	“Set Piece Efficiency”; set-piece tendencies (corners, shots ≤ 20 s) in Opp Scout	Dedicated set-piece shot maps
Possession Recovery	Implemented	“Ball Recoveries by Zone” (zonal heatmap)	High/mid/low zonal comparison
Post-Match Analysis	Implemented	“Post-Match Tactical Summary”, goals vs xG, recoveries, defensive actions	Auto expected-vs-actual narrative
Final-Third Entries	Implemented	“Final Third Entry Analysis (RM)”, “Field Tilt (Final Third %)” card & trend	xG-from-entries, shot-probability maps
Match Info Summary	Implemented	Date, score, opponent, result, venue	Historical scores, contextual info

Beyond the proposal, the live dashboard also ships: GK Distribution Map · Goalmouth Map · Penalty Analysis · Crossing Patterns · Chance Creation Heatmap · Team & Player Radars · Player touch heatmaps · Opponent Threat Heatmap & auto-scout report · a full Benchmarking suite (Goals F/A, Win %, xG F/A, pass accuracy, KPI comparison) · and one-click PDF/DOCX Report export.

8 · Feasibility, Academic Value & Future Scope

8.1 Feasibility & Academic Value

- **Feasible:** uses accessible event-level data; computationally light with Python, Pandas and Streamlit — already deployed publicly via Render.
- **Academic value:** bridges football-analytics theory and practice; every metric is formula-transparent and traceable to observed data, with no synthetic values.

8.2 Future Scope

- **Expanded dataset:** more matches, seasons and competitions for robustness.
- **Player-level analysis:** individual tendencies and contribution to team tactics.
- **Opponent analysis:** automated multi-opponent comparison.

- **Predictive modelling:** forecast xG, transition success, or match outcomes.
- **Interactive expansion:** real-time filtering by player, phase and event type.
- **AI / ML tactical insights:** cluster attack/defence strategies, optimise training.
- **Video integration:** sync event data with footage to validate tactical observations.
- **Multi-competition analysis:** LaLiga, UCL and Copa del Rey in one platform.

Expected impact: a professional-grade analytics platform supporting coaching decisions, player development, and opponent scouting across Real Madrid's competitive calendar.

Real Madrid CF Tactical & Player Performance Analytics · Data: Opta Stats Perform · Python · Pandas · NumPy · Plotly · Streamlit

Appendix · Project Summary Poster

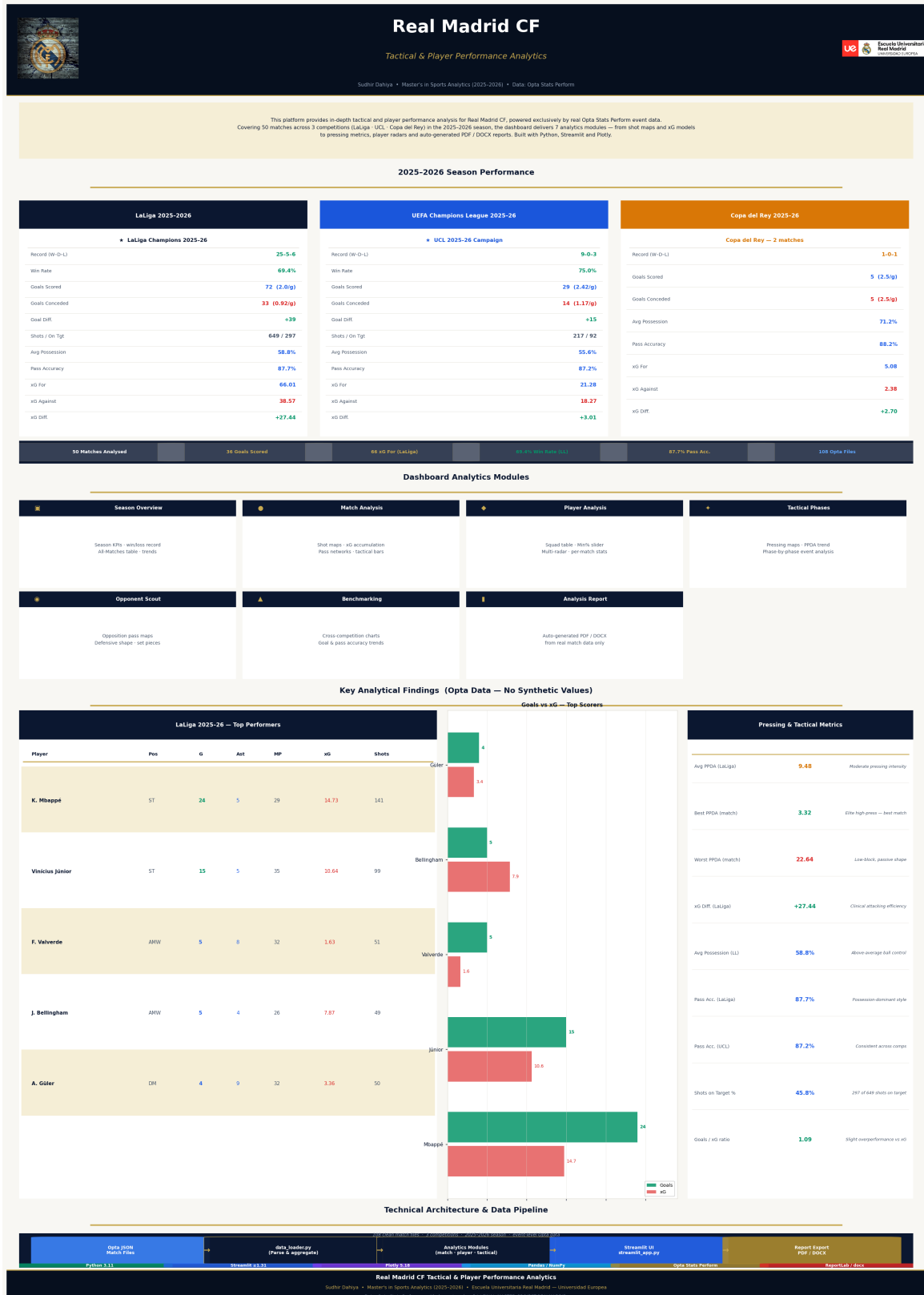


Figure 4. Real Madrid CF Tactical & Player Performance Analytics — one-page project summary.